

WORK SHEET - 21

INDEFINITE & DEFINITE INTEGRATION

[SINGLE CORRECT CHOICE TYPE]

Q.1 to Q.10 has four choices (A), (B), (C), (D) out of which **ONLY ONE** is correct.

Q.1 If $f(x) = \int_0^x (1+t^3)^{-1/2} dt$ and $g(x)$ is the inverse of f , then the value of $\frac{g''(x)}{g^2(x)}$ is

- (A) $\frac{3}{2}$ (B) $\frac{2}{3}$ (C) $\frac{1}{3}$ (D) $\frac{1}{2}$

Q.2 If $\int (x^9 + x^6 + x^3)(2x^6 + 3x^3 + 6)^{\frac{1}{3}} dx = \frac{1}{A} (2x^9 + 3x^6 + 6x^3)^B + C$, where C is integration constant then AB is equal to
(A) 32 (B) 16 (C) 8 (D) 4

Q.3 If $\int e^x (\tan x - x - 2 \tan x \sec^2 x) dx = e^x f(x) + C$ where $f(0) = 0$, then the value of $f\left(\frac{\pi}{4}\right)$ equals
(where C is the constant of integration)

- (A) $\frac{\pi}{4}$ (B) $1 - \frac{\pi}{4}$ (C) $-\frac{\pi}{4}$ (D) $\frac{\pi}{2}$

Q.4 If $f(x) = \left| \sin x - 2\sqrt{2} \cos x + 2 \right| + \frac{1}{1+t^2}$ where $x, t \in \mathbb{R}$ and $g(t)$ is the minimum value of $f(x)$ for $x \in \mathbb{R}$, then the value of $\int_{-\infty}^{\infty} g(t) d(t)$ is

- (A) 1 (B) 2 (C) $\frac{\pi}{4}$ (D) $\frac{\pi}{2}$

[Note: $[k]$ denotes greatest integer less than or equal to k .]

Q.5 $\int_0^{100\pi} \left(\sum_{r=1}^{10} \cot rx \right) dx$ equals

- (A) 0 (B) 100π (C) -100π (D) 100

Q.6 If $f(x) = \begin{cases} x[x] & ; 0 \leq x < 2 \\ (x-1)[x] & ; 2 \leq x \leq 3 \end{cases}$, then $\int_1^3 f(x) dx$ equals

(where $[k]$ denotes greatest integer less than or equal to k .)

- (A) 9 (B) 8 (C) 4 (D) 3

Q.7 The integral $\int \frac{x}{2-x^2+\sqrt{2-x^2}} dx$ equals

(A) $-x \ln \left| 1 - \sqrt{2-x^2} \right| + c$

(B) $\ln \left| 1 + \sqrt{2+x^2} \right| + c$

(C) $-\ln \left| 1 + \sqrt{2-x^2} \right| + c$

(D) $x \ln \left| 1 - \sqrt{2+x^2} \right| + c$

Q.8 If the integral $\int \frac{(\cos 8x + 1)}{(\cot 2x - \tan 2x)} dx = A \cos 8x + k$, where k is an arbitrary constant, then A is equal to

(A) $\frac{1}{8}$

(B) $\frac{1}{16}$

(C) $\frac{-1}{16}$

(D) $\frac{-1}{8}$

Q.9 $\int_{-3}^3 x^8 \{x^{11}\} dx$ is equal to (where $\{ \}$ denotes fractional part of x)

(A) 3^8

(B) 3^7

(C) 3^9

(D) none of these

Q.10 For $n \in \mathbb{N}$, let $P_n = \int_1^e (\ln x)^n dx$, then $(P_{10} - 90P_8)$ is equal to

(A) $10e$

(B) $-9e$

(C) -9

(D) 10

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[PARAGRAPH TYPE]

Q.11 to Q.13 has four choices (A), (B), (C), (D) out of which **ONLY ONE** is correct.

Paragraph for question nos. 11 to 13

Let $f(x) = 4x + \int_0^1 (12x + 20y)xy f(y)dy$

Q.11 Maximum value of $8f(x)$ is equal to

(A) 2

(B) 1

(C) 3

(D) -2

Q.12 Number of point where the function $y = |f(|x|)|$ is non differentiable, is

(A) 1

(B) 2

(C) 3

(D) 4

Q.13 $\int_0^1 [f(x)]dx$ equals, where $[k]$ denotes greatest integer less than or equal to k

(A) $\frac{1}{2}$

(B) 1

(C) $\frac{-1}{2}$

(D) 2



[MULTIPLE CORRECT CHOICE TYPE]

Q.14 to Q.17 has four choices (A), (B), (C), (D) out of which **ONE OR MORE** may be correct.

Q.14 For $I_{m,n} = \int_0^1 x^{n-1} (\log x)^m dx$, let $\sum_{r=1}^{51} \frac{I_{r,52}}{I_{r-1,52}} = k$ then

Note : $[k]$ denote the greatest integer less than or equal to k .

(A) $k = \frac{-51}{2}$ (B) $[k] = -26$ (C) $k = -26$ (D) $[k] = -25$

Q.15 Let f be a non-constant polynomial function satisfying the relation $f(f(x)) = 12x + 10 - f(x)$ for all

$x \in \mathbb{R}$. Then the value of $\int_{-1}^1 f(x) dx$ is equal to

(A) 0 (B) 4 (C) -10 (D) -4

Q.16 Let $f(x)$ be a polynomial of degree less than or equal to 3 with leading coefficient unity, $f(0) = 4$ and

$f(2) = 0$. If $\int \frac{f(x) dx}{x(x+1)^2(x-2)^3}$ is a logarithmic function then

(A) $f(1) = 2$ (B) $f'(2) = 0$ (C) $\int_0^4 f(x) dx = 16$ (D) $f'''(3) = 4$

Q.17 If $\int \frac{\tan x dx}{5 + 3 \tan^2 x} = k \ln |f(x)| + c$, where c is arbitrary constant, then $\frac{1}{k}$ equals

(A) 4 (B) -4 (C) 5 (D) -5

[INTEGER TYPE]

Q.18 to Q.21 are "Integer Type" questions. (The answer to each of the questions are upto **4 digits**)

Q.18 If $A = \int_1^{\tan \theta} \frac{t \, dt}{1+t^2}$, $B = \int_1^{\cot \theta} \frac{dt}{t(1+t^2)}$, then find the value of determinant

$$\begin{vmatrix} A^3 & A^3+5 & B^3 \\ 2^A \cdot 2^B & B^3+8 & -3^A \cdot 3^B \\ B^5 & A^2+B^2-2 & A^5 \end{vmatrix}.$$

Q.19 Let $f(x) = \int_{\frac{1}{a}}^a \frac{\tan^{-1} t}{te^x} dt$, $a \in \mathbb{R}^+$. If $f(0) - f'(0) = \pi / \ln 5$, then find sum of all possible values of a .

Q.20 If $I_n = \int \frac{e^{(n+1)x} dx}{\left(1 + e^x + \frac{e^{2x}}{2!} + \dots + \frac{e^{nx}}{n!}\right)} = \lambda_n (e^x - \ln(f_n(x))) + C$

where $f_n(0) = 1 + \frac{1}{1!} + \frac{1}{2!} + \dots + \frac{1}{n!}$ and C is constant of integration and $g(x) = \lim_{n \rightarrow \infty} \ln(f_n(x))$,

then find the number of real solutions of the equation $g(x) = 4x^2$.

Q.21 If $\int_0^{100\pi} \left(1 - \frac{\sin^2 x}{1 + \cos x} + \frac{1 + \cos x}{\sin x} - \frac{\sin x}{1 - \cos x} \right) dx = k$, then find the value of k .

ANSWER KEY

Q.1	A	Q.2	A	Q.3	C	Q.4	B	Q.5	A
Q.6	C	Q.7	C	Q.8	C	Q.9	B	Q.10	B
Q.11	B	Q.12	C	Q.13	C	Q.14	AB	Q.15	BC
Q.16	ABC	Q.17	ABCD	Q.18	0000	Q.19	5	Q.20	3
Q.21	0200								

